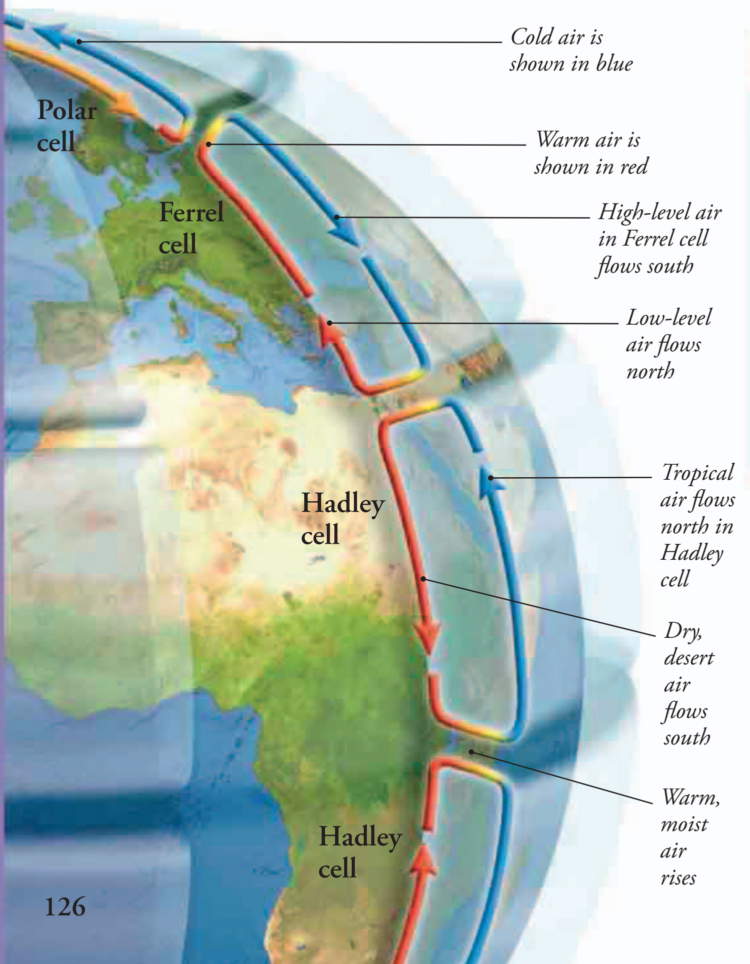
Air currents

The Sun's warmth causes air in the lowest layer of the atmosphere to circulate in currents. But the Sun does not heat Earth's surface evenly. Areas near the equator (the imaginary line that divides the Northern and Southern Hemispheres) receive far more heat than the poles. The warm air from the equator rises and flows north and south, then cools and sinks, creating systems of circulating air.

TROPICAL CIRCULATION

The regions nearest the equator are known as the tropics. Sunlight hits these regions more directly than it strikes the rest of the world, concentrating its energy, which is why the tropics are warmer than the poles. Over the warm oceans, the moist air rises and cools. As it cools, the moisture condenses to form tall storm clouds, which can generate torrential rain. This warmth and rain fuels the growth of lush rainforests.



CONVECTION CELLS

The Sun-warmed ground heats the air above it. This air expands, becoming less dense, and starts to float upward though cooler, denser air—just like the heated air in this paper lantern. Expansion eventually makes the air cool down to the point where it stops rising and starts to sink. This process creates circulating currents of air called convection cells.

CIRCULATING AIR CELLS

As warm air rises, it cools and starts to sink, circulating in a convection cell. In the tropics, air flows north and south in two loops called Hadley cells. Two loops called Polar cells circulate air in the polar regions. There, chilled air sinks down to near ground level, flows back towards the equator, rises again as it warms, and then flows back toward the pole. Each pair of Polar and Hadley cells combines to drive a Ferrel cell, which circulates in the opposite direction.